



THE UNIVERSITY OF SYDNEY
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING

Thesis Literature Review

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1 Introduction

In the modern day with the evolution and widespread adoption of renewable and inverter based distributed energy resources (DERs), there has been a large shift in how distribution networks operate, and how they need to be managed. This adoption has been pushed by the rise of global warming concerns, leading around 145 countries to have or be considering net zero goals, which will result in the complete phase out of coal and gas generators and a massive uptake in renewable DERs [1]. This presents multiple unique problems due to the inherent nature of inverter based DERs and turbine-based generators. The uptake of inverter based DERs is mainly photovoltaics systems (PVs), wind power plants and battery energy storage systems (BESS). Both PVs and batteries have widespread adoption among household owners, leading to the unique problem of overvoltage due to reverse power flows in distribution networks during periods of high generation [2]. Frequency issues have also been increasing as the inverter-based power sources do not provide inertia that turbine generators do. Inertia is a natural part of turbine-based energy sources, which helps absorb small frequency disturbances by slowing the rate of change of frequency. Slowing the rate of change gives time for the grid control systems to activate [3]. Without this inertia, frequency disturbances become much more of an issue, which is why the dynamic virtual power plant is needed [4]. A virtual power plant is an aggregation of distributed energy resources, controlled together to act as a power plant [5].

2 Background

2.1 Virtual Power Plant Background

A virtual power plant is an aggregation of distributed energy resources, controlled together to act as a power plant [5]. A VPP traditionally provides economic dispatch, using setpoints as the main control method. However VPPs have evolved to be optimisation based, with programmed responses to network deviations [6]. Yet this optimisation does not take into account each devices parameters in terms of speed, leaving it ill-equipped for fast frequency response [7].

2.2

The study of DVPPs is new and cutting edge, as can be observed through publication data. In figure 1 [5] the number of publications per year for a virtual power plant can be observed according to Scopus, with the first few papers coming in the late twentieth century, before a massive rise to in publications in the late 2010s and early 2020s. The number of papers peaked in 2022 with 1,227 papers declining after this year likely due to the saturation of publications in previous years.

2.3

The earliest paper published on IEEE on virtual power plants was published in 2003 and displays how the concept of a virtual power plant has evolved [8]. Although the idea of aggregating distributed energy resources is the same, what these resources are and how they should be implemented is drastically different. The VPP in [8] assumes the devices are mostly rotating and turbine devices, which is drastically different to the modern VPP, made of mostly inverter-based PV and batteries. The control method was also very different, following more traditional methods of setpoints based on predictions. The virtual power plant would be assigned based on forecasts an amount it should produce over a period, usually between 5 minutes and an hour. Modern virtual power plants build on this by also using closed loop control schemes, adjusting output depending on measured values such as frequency. Grid stability is not even mentioned in the paper, which is a core issue in discussed in modern VPPs.

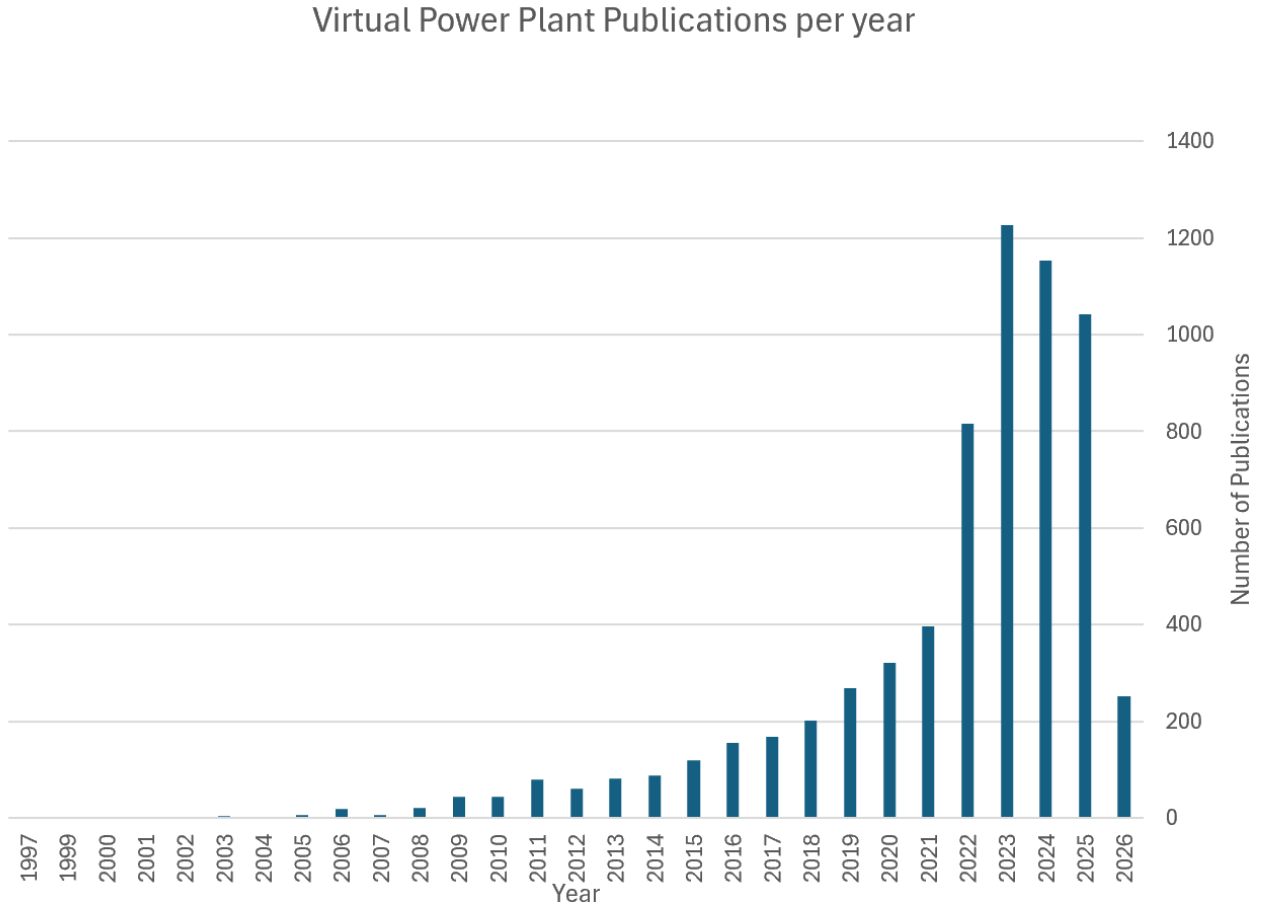


Figure 1: Virtual Power Plant Publications per year [9]

2.4

The dynamic virtual power plant publications show a different trend to VPPs, with the largest number of papers being published in 2025 with only 13 papers. The general trend observed in figure [10], can be seen that the number of papers published is slowly increasing since the first paper published on according to Scopus in 2012. However these first papers had a different definition of a DVPP than the moder definition, with the first paper on the modern definition being published in 2022, as observed in figure 3. Particularly in the 2020s there has been an increase in papers, although the total number of papers is still low, displaying the cutting-edge nature of DVPPs as a concept.

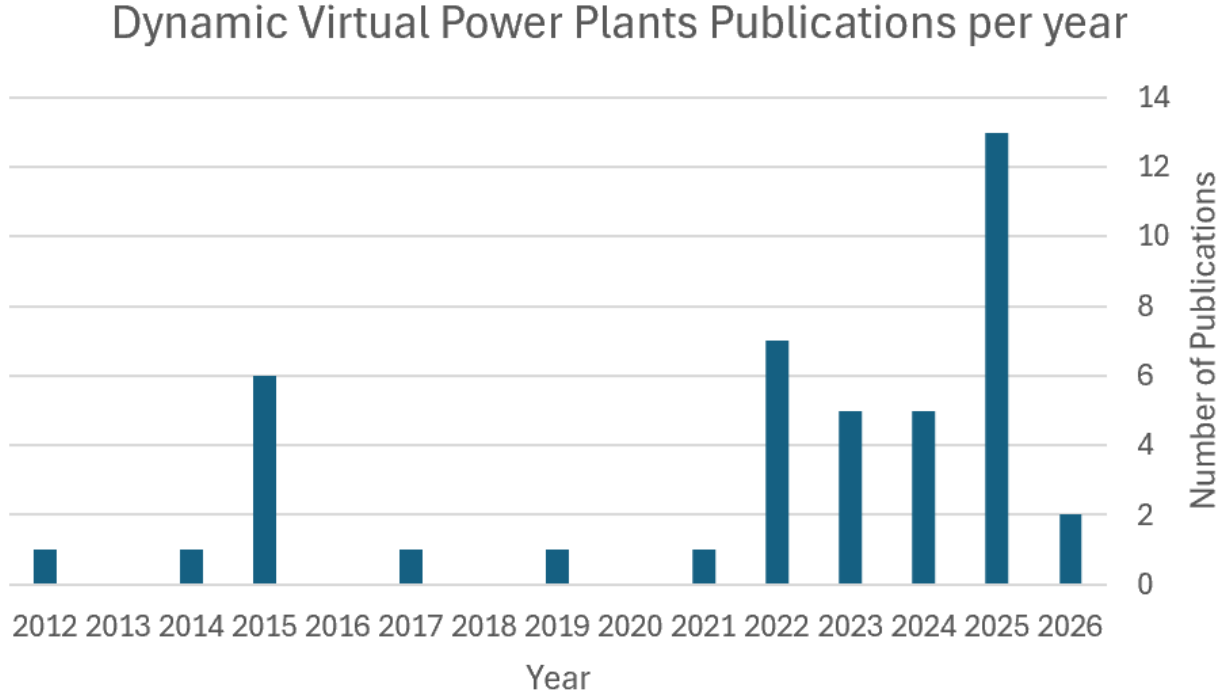


Figure 2: Dynamic Virtual Power Plant Papers per year unfiltered
[10]

2.5

The first paper on IEEE that I could find mentioning dynamic virtual power plants was published in 2012 and is much more advanced than the first paper on VPPs [11]. However, it defines a dynamic virtual power plant as a virtual power plant in which devices can dynamically join and leave, which is not the modern definition. It does, unlike the 2003 paper, discuss grid stability and a central controller as issues. The control system is still based on predictions, not using the modern closed loop system. The modern DVPP was defined in the 2022 paper [12] where it is stated that a "A DVPP is a set of dispatchable and non-dispatchable RES along with a set of common control and operation procedures." [12]. It further defines a DVPP as a DVPP is "concerned with dynamic ancillary services, such as fast frequency and voltage control." [12] and that a DVPP must act dynamically both "locally (for each RES generator), globally (for grid ancillary services and interaction with other close-by elements of the grid) and economically (for internal optimal dispatch and participation in electricity markets) [12]". Acting dynamically means the response of the system changes continuously based on measured inputs, unlike statically where the system acts with a defined response once a specific input condition is met [10].

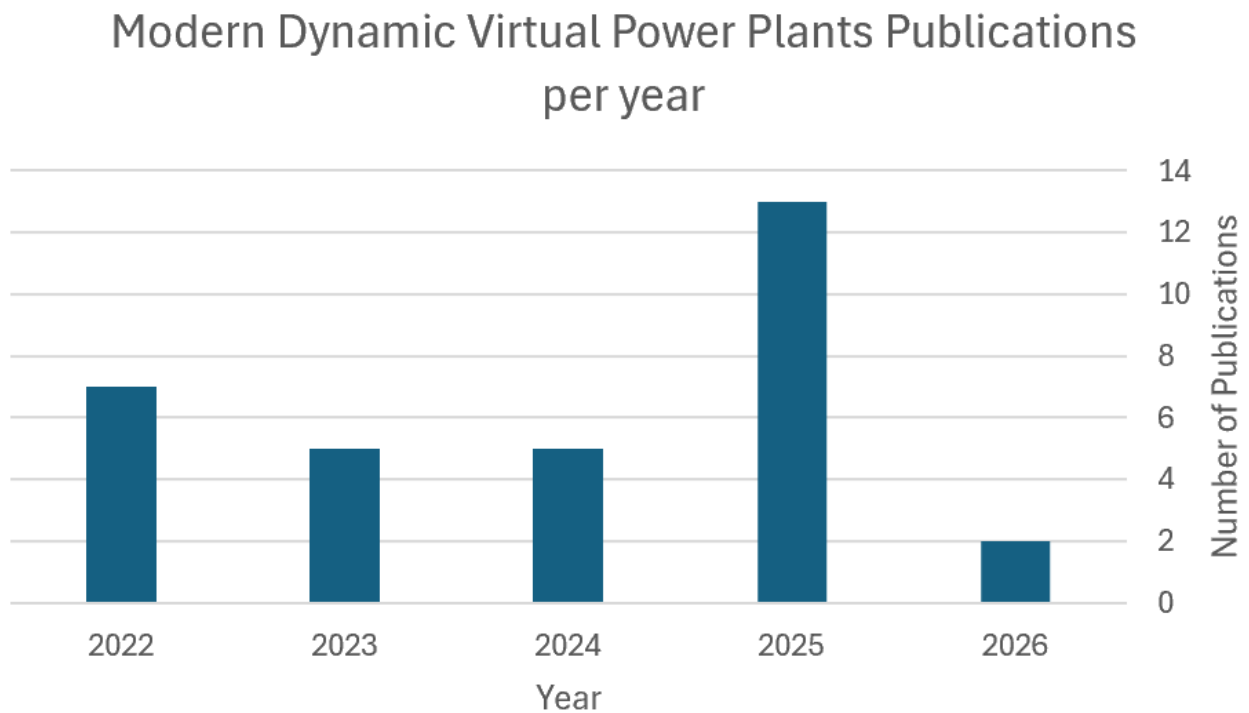


Figure 3: Dynamic Virtual Power Plant Papers per year filtered [10]

2.6 First modern DVPP paper

The paper "Dynamic virtual power plant: A new concept for grid integration of renewable energy sources provides important context for DVPPs as it explains the fundamentals of DVPPs and why they are important. The rise of renewables in Australia, can be seen in figures [Figure 4](#) and [Figure 5](#), showing the cumulative PV capacity increase in Australia as well as the cumulative battery installations.

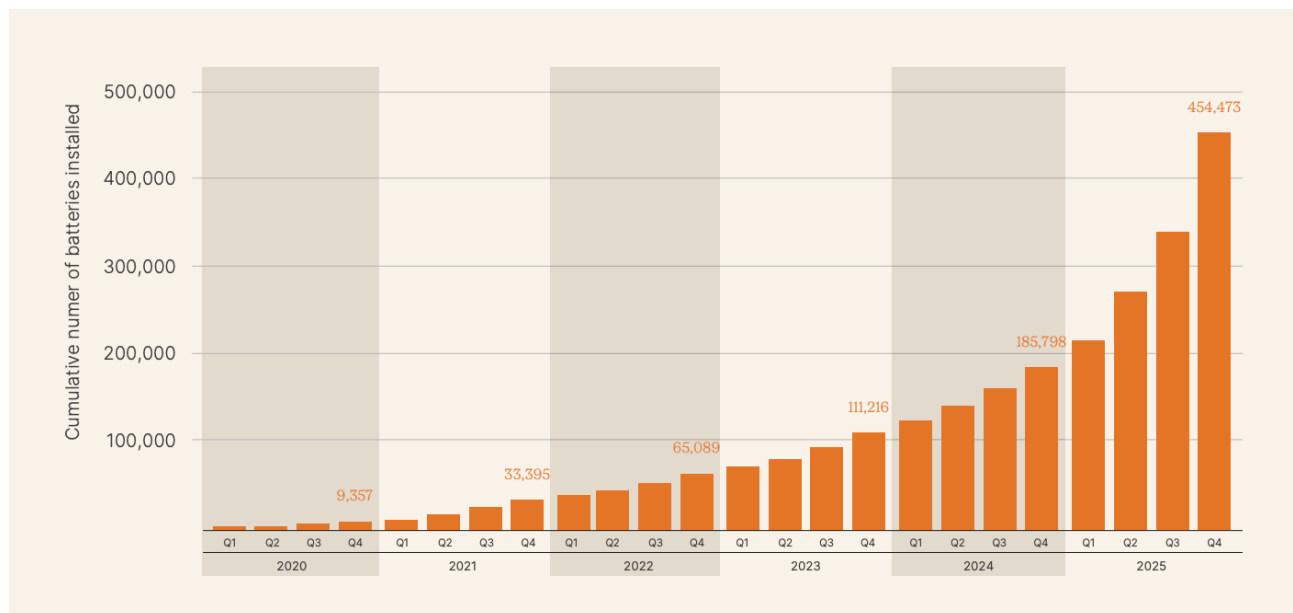


Figure 4: Dynamic Virtual Power Plant Papers per year filtered

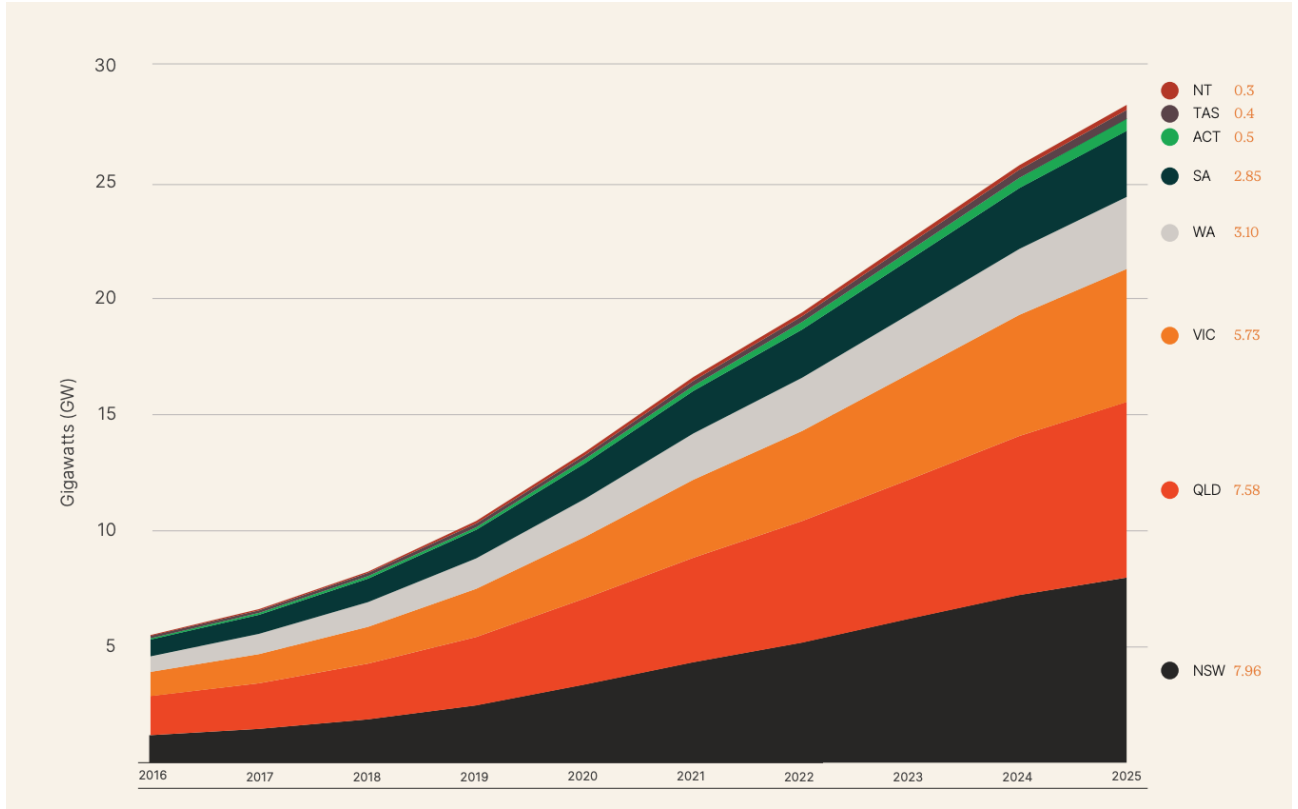


Figure 5: Dynamic Virtual Power Plant Papers per year filtered

This has led to the need of shifting ancillary services from standard generators to dynamic virtual power plants. However the implementation of a DVPP is much more complicated, due to its distributed and nonhomogeneous nature. The nonhomogeneous nature of a DVPP means that it is made up of lots of different devices with different response times, power outputs as well as being dispatchable and non-dispatchable. To deal with the distributed and nonhomogeneous aspects, the paper proposes that a DVPP must be managed on three levels, local, global and economic. The local level of control involves controlling each individual device. Each separate device must be operated such that it doesn't exceed its limits, and that it follows the instructions for power output it is given. This is particularly important as the DVPP must factor in the constraints of the DERs, in terms of immediate power output, and the level of control over this power. Solar and wind may need large changes to their local controllers, as the standard way they operate is called maximum power point tracking (MPPT), which means the device is operating at its maximum output at all times. This conflicts with providing ancillary services however, as this would require an instantaneous increase in power, not possible with MPPT. Therefore the pv and wind controllers must be changed so that they keep reserve power to participate in the DVPP ancillary services. The global level is that the DVPP must be controlled such that all the DERs when aggregated together have the desired outcome. This means that the DVPP must act like a generator, including providing a virtual inertia as it does not have the spinning machines of a generator. Providing these ancillary services is not an easy process, due to the spatially distributed aspect of a DVPP, and that DERs are designed with controllers to act as a single entity, not as a part of a DVPP. The final level is the economic level, in which a DVPP must participate in energy and ancillary markets as if it were a generator. This can be difficult for the energy markets, due to the scheduled nature of them, and the unpredictability of PV and wind power output. This means that the DERs have to be regularly redispatched to keep the commitment made to the energy market.

2.7

To integrate a DVPP successfully a precise control system must be implemented. The key difference between a VPP and a DVPP is that a DVPP provides dynamic ancillary services, which is fast frequency and voltage control [12]. VPPs can also provide ancillary services but do this through slow response static ancillary services, which include injections of baseline power, as well as set-point tracking, in which a centralised point will track the frequency and voltage of the system and send out instructions for how the DERs should act based on the

grids state compared to the setpoint (desired state).

2.8 Adaptive Divide and Conquer

A method [13] came up with for optimising DVPPs was to use an adaptive divide and conquer approach. This method works by first determining the response desired from the DVPP to some type of disturbance. This desired response is formulated as a diagonal Multiple Input Multiple Output (MIMO) transfer matrix. This transfer matrix when multiplied by the measured frequency and voltage deviation outputs the desired active and reactive power output from the DVPP. The desired active power is determined by the change in frequency, and the desired reactive power is determined by the change in voltage. This decoupled f-p, v-q method is regularly used for transmission systems, as they have almost negligible resistance when compared to inductance, however this can cause issues in distribution networks where resistances are higher. Once this MIMO desired transfer matrix is calculated, it is divided among the DERs of the DVPP using adaptive dynamic participation matrices (ADPMs). These ADPMs are essentially frequency filters, telling the device they are assigned to how quickly it should respond. Quick response devices such as batteries are assigned high pass filters, slow response energy sources such as hydro are assigned low pass filters, while devices in the middle are provided band pass filters. The power output of each of these devices is controlled by the DC gain of the ADPMs, which altogether should sum to the desired output. Each of the devices in the DVPP has a controller that makes the device match the desired output as closely as possible without exceeding device limits. This controller is designed such that it knows the optimal response to the boundary conditions of the devices range. These responses are blended to get the desired output based on the current value of the ADPM DC gain values.

2.9

Paper [14] implements a similar control method to paper [13], using dynamic participation factors to assign what they should contribute to the DVPP. It is used similarly to paper [13] as these factors are assigned based on the dynamics of the device, slow devices such as hydro are assigned the slow recovery part of the response, whereas fast devices such as wind and batteries are assigned the instant response. The way the paper differs however is in how the participation factors are designed, as well as the complexity of implementation. The participation factors are calculated via a simple formula based on the device, limiting the devices compared to paper [13]. This results in a scalar output, unlike the matrix in paper [13]. The paper is also built on the fact that the devices are non-minimum phase, meaning that hydro initially act opposite to instruction and wind initially responds correctly and then acts opposite. The controller of each device is also much simpler, not updating participation factors online depending on available power. While paper [14] method for optimisation is strong, it is limited to the system it is built on.

A core component which allowed the model implemented by Bjork to be effective was the model that was designed so that wind turbines could effectively participate in fast frequency response[15]. This was designed by the same team who made the nordic model, and addresses the problem mentioned earlier with PV and wind needing to keep reserve power so they can participate in ancillary services[12]. To do this [15] detail a method in which the wind power plant provides FFR while still operating at the MPP, without any curtailment. This must be done as if the wind turbine needs to provide power whilst running at the MPP, it does so by using kinetic energy, which slows down the wind turbine. This slow down causes the wind turbine to enter an unstable operating region, which the turbine must then recover from. This leads to the wind turbine being unable to provide ancillary services during this time. The solution to this is to allow the wind turbine to stably run at a speed lower than the MPP, when power is needed, and then return back to the MPP. This isn't easy to implement however as this speed will be in the unstable operating region. One of the contribution [15] provides is a controller which allows the wind turbine to operate in this unstable condition. This controller is made using a closed loop feedback system in which the equation for output power of the wind turbine is

$$P_e = G(s) \frac{sM(\Omega_0) - R(\Omega_0)}{sM(\Omega_0) + G(s)\hat{K}(s) - R(\Omega_0)} P_{\text{ref}} \quad (2.1)$$

When the wind turbine operates at a point below optimal speed, $R(\Omega_0)$ becomes positive, meaning that the turbine is not in a stable operating condition. Equation 2.1 manages to stabilise the turbine in this unstable operating position, but in the process introduces an NMP zero. Therefore to include this NMP into the DVPP algorithm, the equation 2.2 is used to represent the NMP.

$$H_{\text{wind}}(s) = \frac{s - \bar{z}}{s + \bar{z}} \quad (2.2)$$

These equations are used to model a wind turbine which can effectively participate in ancillary services and is

used in the model [14] made.

2.10 Secondary Frequency Control DVPP

A topic not commonly addressed in DVPPs is the secondary frequency response. The paper [16] addresses this by proposing including a DVPP participation factor in Automatic Generation Control (AGC) just as any other generator would be assigned. For this to work, the DVPP must supply less than its total power at all times, allowing for internal redispatch if one of the DERs output reduces. This redistribution is implemented using a low pass filter with a four second time constant. This ensures that it doesn't interfere with primary frequency control which could cause instability.

2.11

A problem not addressed by either [14] or [13] is how to implement the control systems across a spatially distributed grid. [17] explains the issue by describing areas in the grid where power comes from as clusters. If an electrical stability issue occurs on the grid, the closer the cluster is to the issue, the larger the issue may seem. This can create difficulty in assigning the expected response from each individual cluster. This could be done by having an elaborate model of the grid and devices, but this would be incredibly impractical, especially considering distribution networks often do not have detailed models on them[18]. Instead [17] proposes measuring frequency and power at each individual cluster and assigning response based off of these measurements. This response is assigned using equation 2.3, where $2H_{\text{DVPP}}^i$ represents the emulated inertia constant and $P_L^{\text{cluster}_i}$ represents the power imbalance for a cluster in the DVPP.

$$2H_{\text{DVPP}}^i \frac{d^2 \delta_i}{dt^2} = T_{m,i} - T_{e,i} = \Delta P_L^{\text{cluster}_i} \quad (2.3)$$

This equation is then simplified down to equation ?, where RoCoF represents the rate of change of the frequency.

$$2H_{\text{DVPP}}^i \times \text{RoCoF}_i = \Delta P_L^{\text{cluster}_i} \quad (2.4)$$

The inertia constant is then calculated using the measured frequency power relationship at each individual clusters connection point. This is then used to find the contribution factor of each cluster, by taking the maximum of two calculated contribution factors. One is the steady state contribution factor, calculated by the desired frequency response to return to the nominal frequency, after the initial drop is prevented. The other contribution factor is the dynamic one, which is based off of the RoCoF. These contribution factors are then used with the frequency reserves to provide the desired response.

2.12 Grid Forming

As mentioned earlier, a problem [13] didn't address in their first paper was the distributed nature of the DVPP. Therefore they have a follow up paper in which they address this problem, while also proposing a method in which DVPPs could become grid forming[19]. The term grid-forming according to AEMO [20] "inverters set their own internal voltage waveform reference and can synchronise with the grid or operate independently of other generation." This is different to standard, as inverter based DERs normally just synchronise to a grid which is formed by large generators. Tayyebi [21] demonstrated that while inverter-based DERs can operate in a grid-forming mode, their fast dynamics can destabilise slower synchronous machines when there is a high penetration of inverted-based DERs. To implement grid-forming control while accounting for these risks, Häberle [19] uses the following equations.

$$\Delta f_i(s) = -T_i^{pf}(s) \Delta p_i(s), \quad \forall i \in \mathcal{F}_{\text{orm}} \quad (2.5)$$

$$\Delta q_i(s) = -T_i^{qv}(s) \Delta v_{\text{pcc}}(s), \quad \forall i \in \mathcal{F}_{\text{orm}} \quad (2.6)$$

$T(s)$ is a transfer function, which will give the desired response based on the input. The equation above represents the local level controller, where $\Delta p_i(s)$ and $\Delta v_{\text{pcc}}(s)$ represent the individual active power and voltage deviations at each device. The transfer function from these inputs gives an output frequency and reactive power deviation. These aggregate together to give equation 2.7 below. This is notably different from [13] as instead of receiving a change in frequency and voltage, and imposing a change on active and reactive power to correct this, [19] receives a change in active and reactive power and imposes a change on frequency and voltage. This is due to the grid-forming setup.

$$\begin{bmatrix} \Delta f_{\text{pcc}}(s) \\ \Delta v_{\text{pcc}}(s) \end{bmatrix} = \underbrace{\begin{bmatrix} T_{\text{des}}^{pf}(s) & 0 \\ 0 & T_{\text{des}}^{qv}(s) \end{bmatrix}}_{= T_{\text{des}}(s)} \begin{bmatrix} \Delta p_{\text{pcc}}(s) \\ \Delta q_{\text{pcc}}(s) \end{bmatrix} \quad (2.7)$$

2.13 Spatially Distributed

Paper [19] also makes an important contribution to applying a DVPP control system to a spatially distributed grid. [19] solution uses equation 2.8 to include the line impedance.

$$\begin{bmatrix} p^* \\ q^* \end{bmatrix} = \mathbf{R} \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} \frac{X}{Z} & -\frac{R}{Z} \\ \frac{R}{Z} & \frac{X}{Z} \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix}, \quad Z = \sqrt{R^2 + X^2} \quad (2.8)$$

The problem with implementing [13] adaptive divide and conquer in a spatially distributed system is that the control system assumes a decoupled relationship between active power and frequency, as well as reactive power and voltage. This functions in high reactance networks, such as transmission networks, but falls apart in medium and low voltage networks due to the high $\frac{R}{X}$ ratio. So to account for this [19] introduces equation 2.8, which uses line resistance and reactance to create the decoupled f-p* and v-q* relationship. This solution relies significantly on the fact that the $\frac{R}{X}$ ratio is the same throughout the network, and that this is known, which may not be the case.

3 Battery Background

Batteries are not static power sources. Overtime they degrade, with the two main places they degrade being capacity fade, the amount of charge a battery can hold, and power fade, the amount the battery can discharge or charge at any points in time. This is important to DVPPs as BESS's will be a core component of every modern DVPP and having knowledge on how it can be used is of utmost importance.

3.1 Frequency regulation not impactful

DVPP operators have less need to be cautionary when using BESS then energy arbitrage as frequency regulation is low impact on a battery's SoH. This was recorded in [22] where a year long battery degradation profile for a battery providing frequency regulation observed was mapped to a total battery lifetime of between 8.5 and 13.5 years. This variance depended on how the battery recharged after usage. This is further verified by [22] when comparing to actual batteries used in FR where a 1.6 MW / 0.4 MWh system delivering primary frequency regulation saw a 2% capacity drop over 1.5 years with approximately 390 full cycles. This is due to frequency regulation not requiring deep battery discharge, as is often required from other uses of batteries such as energy arbitrage. Why this positively affects a FR batteries SoH can be observed in [23] where different cycling conditions are compared. When a battery is cycled such that it reaches full charge and complete discharge, this stresses the battery further than when the battery is held at a central charge. In the high and low charge regions solid-electrolyte-interphase growth, lithium plating and cathode structural degradation are accelerated, leading to a quicker degradation in the batteries SoH.

3.2 Calender aging

Batteries in DVPPs will still have significant SoH reductions due to both calender aging and energy arbitrage. As frequency regulation is not constantly needed, batteries can be used both for energy arbitrage and frequency regulation, leading to the batteries SoH decreasing at a standard rate for an energy arbitrage battery. Even if the battery is only used for the DVPP, calender aging has a large impact on the SoH of the battery as seen in .

3.3 SoH degradation affects both charge capacity and power

3.4

The cause of power and capacity fade is usage over time, however tracking the exact values of current SoP and SoH is difficult. Battery health is often tracked with Lithium-Ion Battery Equivalent Circuit Models including the Rint model, the RC model, the Thevenin model, the PGNV model and the DP model. but the most applicable for modern DVPP systems is the Rint-RC hybrid ECM. Rint-RC hybrid ECM combines both more simple but less accurate internal resistance models and more accurate but significantly more complex electrochemical models.

4.2 Hydro

The first power source I implemented into my model was hydro, and with it came the biggest issue with my current model. I used a simscape electrical synchronous machine to represent the Hydro generator, and connected it to a hydraulic turbine and governor to control it. This became the biggest issue of my initial simulation as this hydraulic turbine and governor was not connected to my DVPP control system. This led to the hydro governor having a very large response to frequency deviations, while my DVPP control system had a very small response, due to the lack of communication. The synchronous machine values were chosen to be a 30 MVA rating and a 3 second inertia constant. These values were chosen to try and reduce the impact of the hydro system, so the DVPP control system could be observed more clearly. The governor values were based off of [14], with a water time constant of 2.67, permanent droop of 0.20 and servo time constant was set to 0.5. The NMP zeroes addressed in [14] are implemented by default into the hydro governor, meaning no by hand implementation needed. The plan going forward for hydro is to implement it into the DVPP control system, to create a more coordinated response.

4.3 Battery Energy Storage System

Implementing a BESS into the DVPP control system came with many difficulties, due to an actual simscape battery not being usable for this task. Therefore it's response was instead modelled using the transfer function

$$H_{\text{bess}}(s) = \frac{1}{0.04s + 1} \quad (4.1)$$

which represented the quick response of a BESS system. It was decided for this initial test that the state of charge (SoC) of the battery would not be tracked. I decided this as I wasn't testing economic dispatch, just ancillary services, so the battery was based off of the assumption that its local controller always kept the BESS with enough energy to respond, but also enough space to absorb when needed. The battery is rated at 30 MW and can absorb or output this amount of power. The plan for the BESS going forward is to implement a capacity, SoC and state of health of the individual batteries in the system.

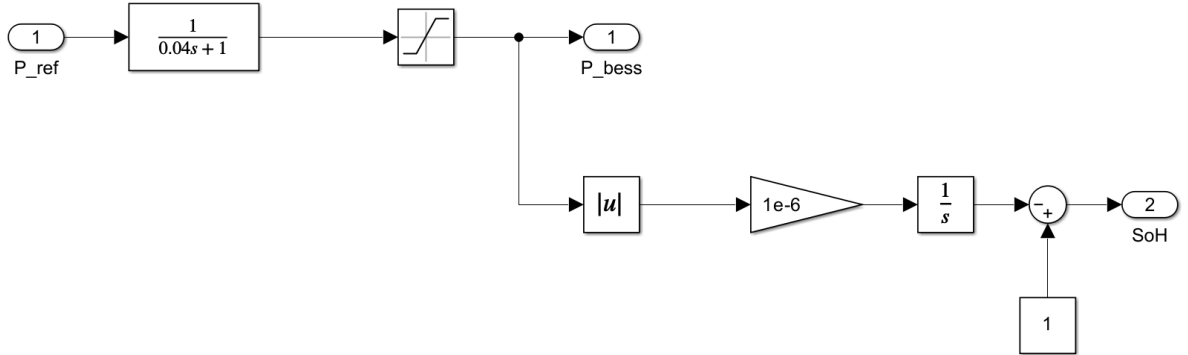


Figure 7: DVPP BESS Power Source

4.4 Wind

The wind power subsystem was designed following the principles explained in [15]. From this paper the NMP transfer function was implemented.

$$H_{\text{wind}}(s) = \frac{s - \bar{z}}{s + \bar{z}}, \quad \bar{z} = 5.8v \times 10^{-3} \quad (4.2)$$

Assuming a wind speed of 8 m/s gives a value $\bar{z} = 0.0464$ which is the value I used in the transfer function. It was decided to just assume a constant value for wind, as implementing a changing value was too difficult for the initial test, but is part of the plan for the next stage. The wind power subsystem is rated at 20 MW, with no ability to absorb power like the BESS, as wind has no storage component. The reason the NMP transfer function is important is it includes the fact that increasing power output from wind slows down the wind turbines, which if not accounted for can lead to the turbines becoming unstable and dropping severely in power.

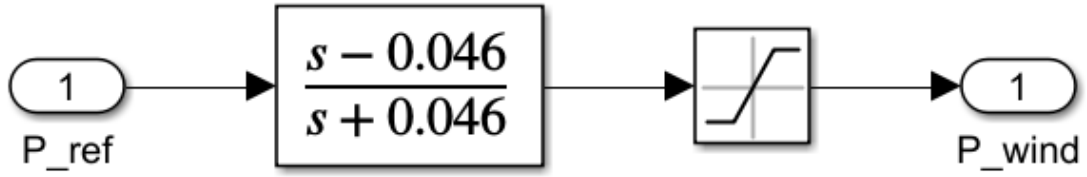


Figure 8: Wind Power Source

4.5 Disturbance

The frequency disturbance for the DVPP to respond to is caused by a three phase dynamic load attached to the network, which at 10 seconds goes from a load of zero to 50 MW. This causes an frequency drop as there is now more load than generation, and similar to how the wind turbines slow down when more power is required, a frequency drop occurs.

4.6 DVPP control strategy

As my goal was to be able to test multiple control structures, the DVPP controller was setup in a variant subsystem. This subsystem would have multiple subsystems inside, which would be selected using the workspace variable DVPP_mode. In this subsystem I replicated two strategies. The DVPP controller recieved two inputs, but only one would be used for the control strategies implemented, which which be negative change in frequency $-\Delta f$. It is standard in ancillary services to use Δf as the input as the response is only based on the change from the nominal value. The DVPP output is the desired power output from the wind and BESS generators. Hydro is not included in this output currently as explained earlier, the Hydro acts independently.

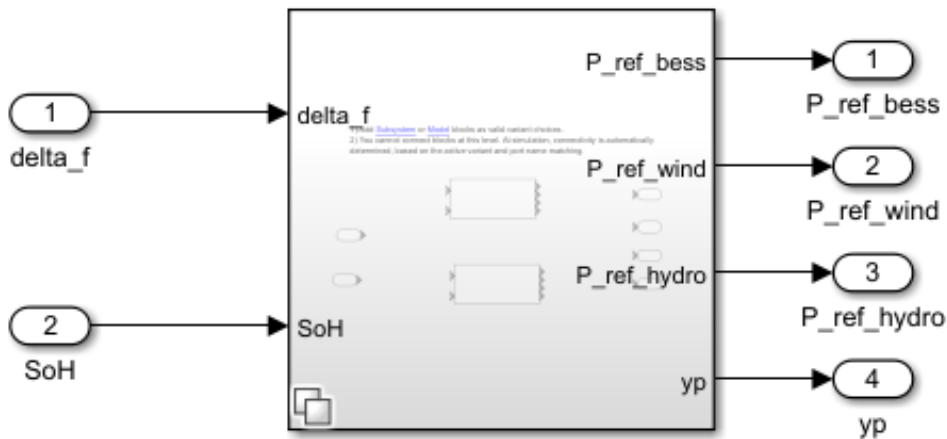


Figure 9: DVPP Controller

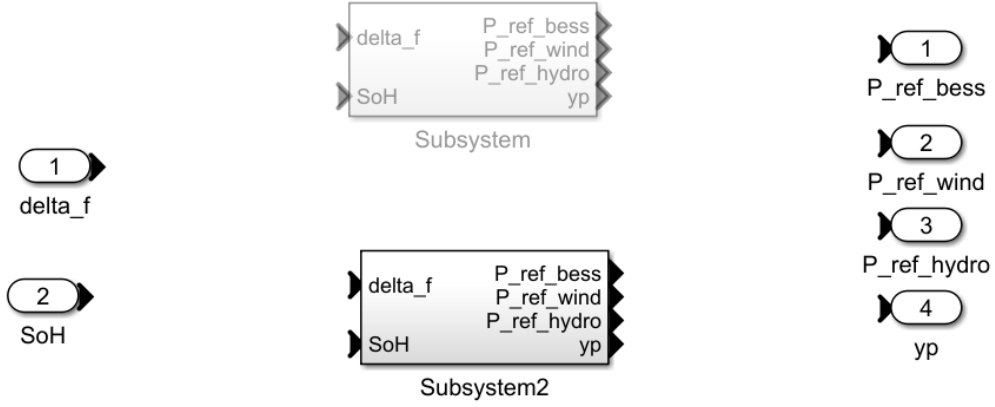


Figure 10: DVPP Controller Systems

4.7 Strategy One: Basic Droop Control

Strategy one implements a standard VPP control strategy, with basic droop control. This means a proportional relationship between Δf and the wind and BESS desired output power is created using gain blocks. The gain values for wind and BESS were calculated using a standard 5% droop characteristic, as observed below.

$$K_{\text{bess}} = -\frac{P_{\text{bess, rated}}}{0.05 \times f_{\text{nom}}} = -12 \text{ MW/Hz} \quad (4.3)$$

$$K_{\text{wind}} = -\frac{P_{\text{wind, rated}}}{0.05 \times f_{\text{nom}}} = -8 \text{ MW/Hz} \quad (4.4)$$

This is a very uncontrolled strategy, as there is no regard for NMP characteristics, or the safety of the power source.

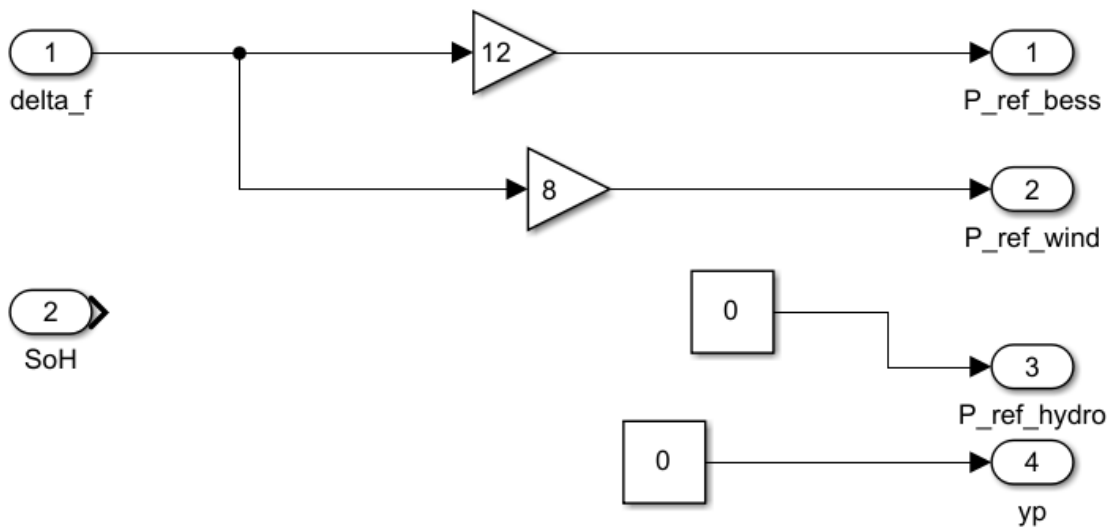


Figure 11: DVPP Controller Strategy 1

4.8 Strategy Two: Nordic Participation Factors

The Nordic Participation strategy, based off of [14] and [15]. Its participation factors are constructed such that the aggregate output of the BESS, Wind and Hydro added up to the desired frequency response, which was decided to be

$$F_{\text{FCR}}(s) = 20 \cdot \frac{4s + 1}{(s + 1)(15s + 1)} \text{ MW/Hz} \quad (4.5)$$

. BESS was allocated 25% wind was allocated 15% and hydro was allocated 60%. While this was a good plan, it didn't work out as hydro is not communicating to the DVPP, as will be explained in the results. These allocations were designed into the controllers through transfer functions. These were implemented through local controller representations where

$$K_i(s) = c_i(s) \cdot \frac{F_{\text{FCR}}(s)}{H_i(s)} \quad (4.6)$$

represents all local controllers. $H_i(s)$ is the dynamic model of the device, which represents how well it can respond quickly. $c_i(s)$ represents the participation factor. BESS is given $H_i(s) = 1$ as it is incredibly good at quick response, whereas wind is given $H_{\text{wind}}(s) = (s - 0.046)/(s + 0.046)$, due to its NMP zero. Due to the complexity of the $K_i(s)$ result, the transfer functions are simplified down third order transfer functions, while still maintaining the response. The resulting transfer functions are in Figure 12.

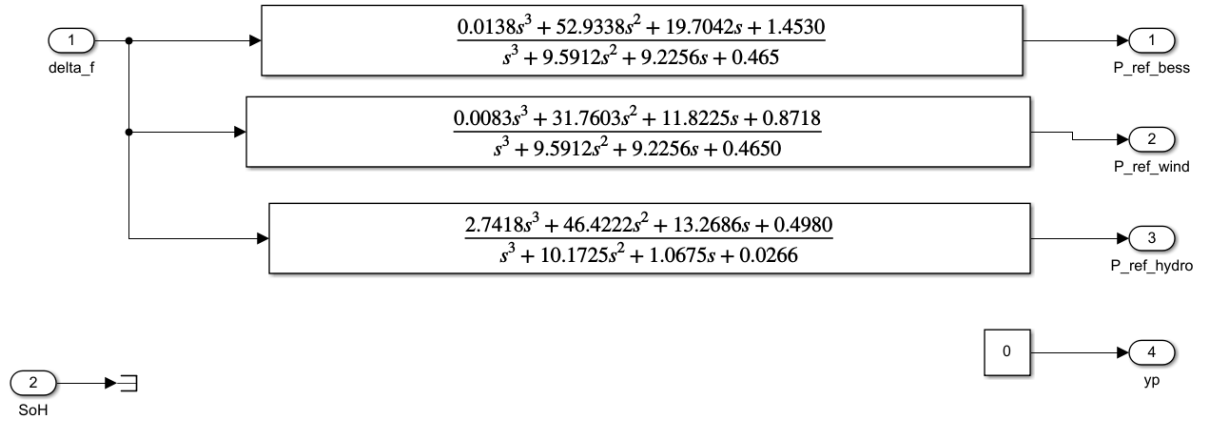


Figure 12: DVPP Controller Strategy 2

4.9 Results

As can be observed in Figures 13 and 18 the results for strategy one and two are very similar, with the only difference being the BESS and Wind response. The Frequency deviation from the added load is the same for strategy one and two, as expected, however the recovery to nominal is also the same, which is slightly unexpected. The reason this occurs is due to the Hydro being able to competently handle the frequency deviation by itself. I tried to reduce its effect by lowering the power output from 100 MVA to 30 MVA, but the hydro is still extremely dominant. This is due to the governor making the hydro act on its own, as well as the fact that the system only has a single bus, making the hydro's inertia very effective.

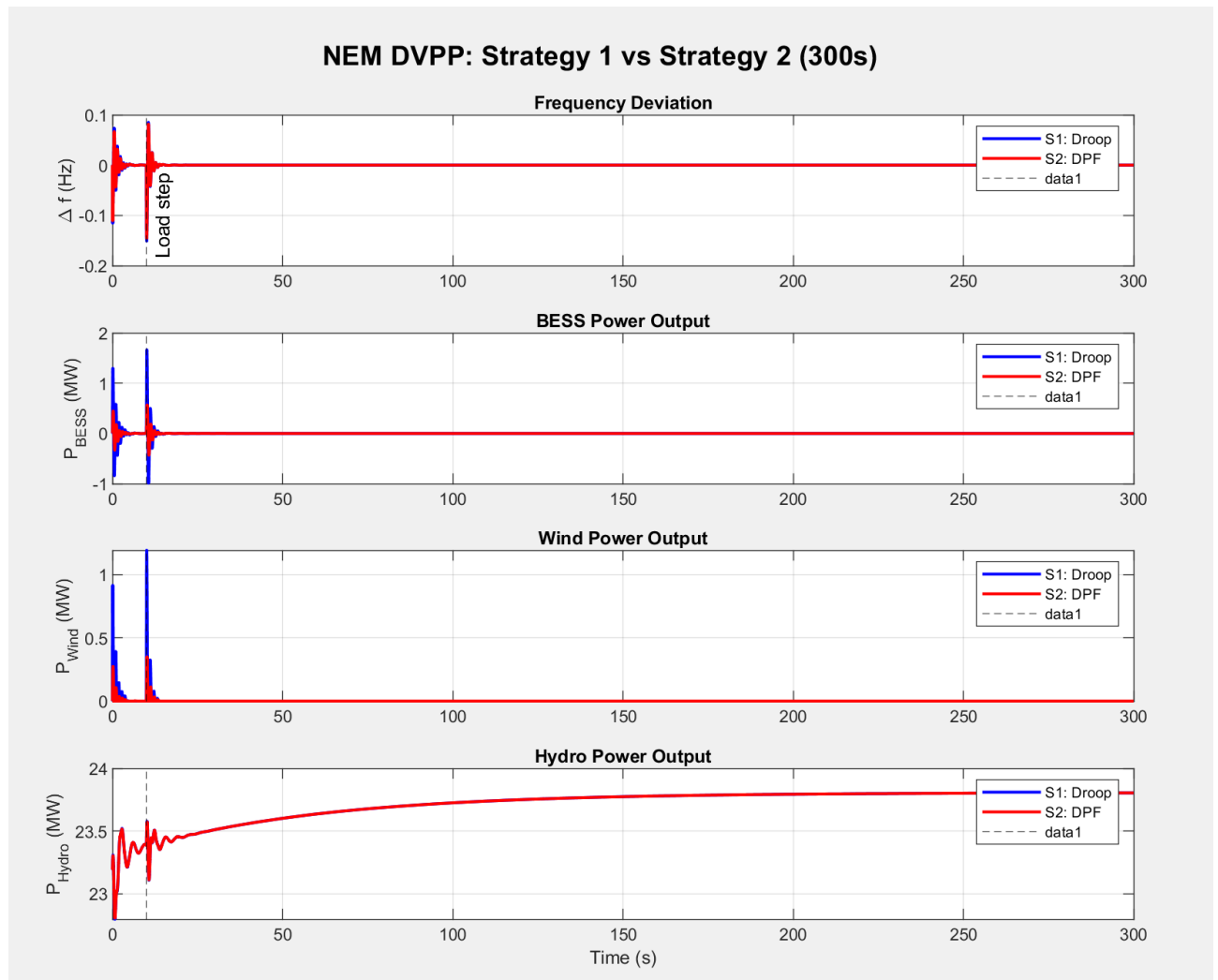


Figure 13: DVPP Results 300 seconds

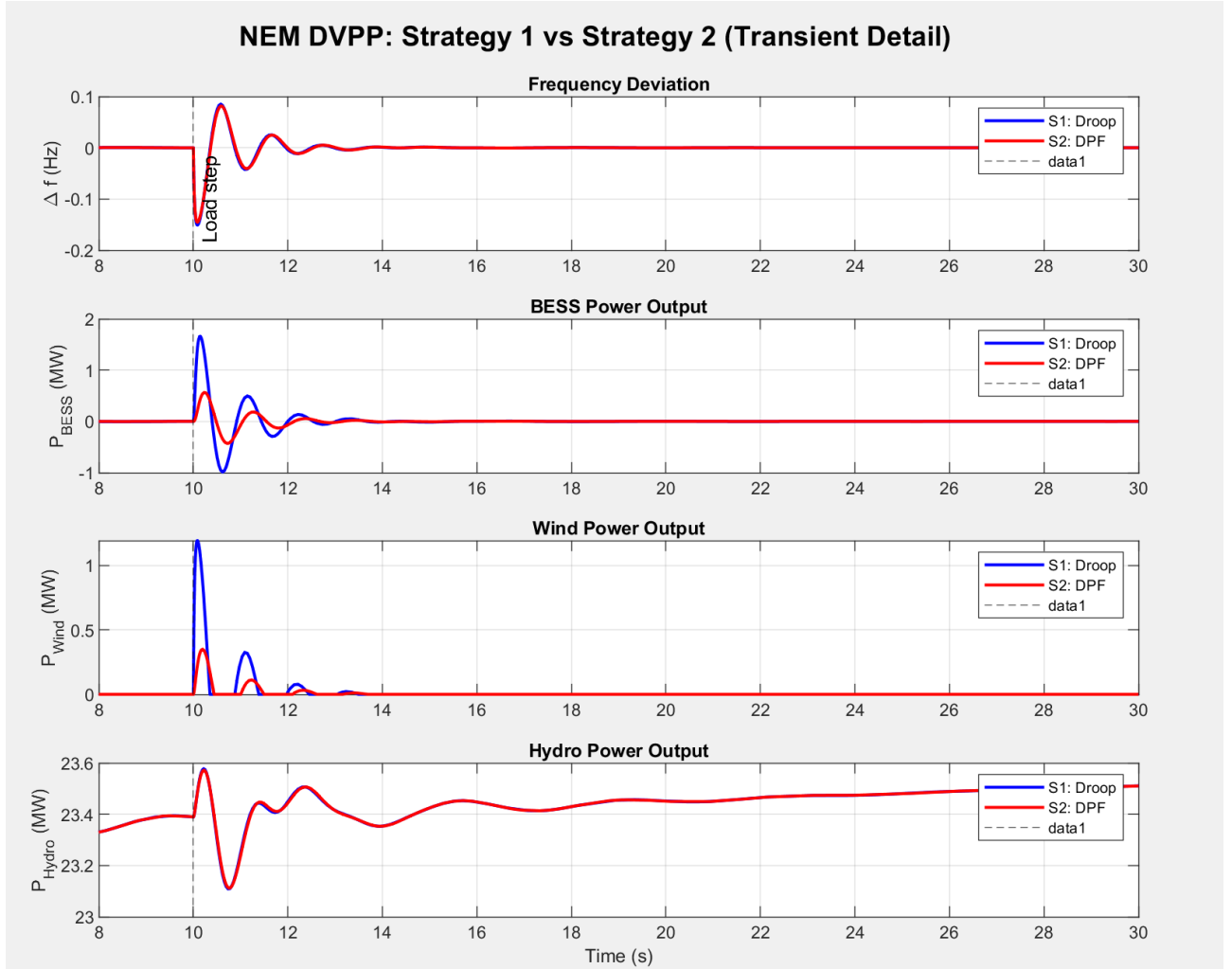


Figure 14: DVPP Results 30 seconds

4.10 BESS response

The BESS response is very similar between the two strategies, with an initial power increase to raise the frequency, the frequency then goes above nominal due to the governor and DVPPs response, resulting in the BESS absorbing some power to help bring the frequency back down, this pattern then occurs again but with a smaller output until the frequency is stabilized. The main difference between the two strategies is strategy two is more controlled and smaller, due to the participation factors allocating 25% of the response to the BESS.

4.11 Wind response

The Wind response is again very similar between the two strategies, and similar to the BESS response, with the main difference being the power output is smaller due to only a 15% allocation. There is also no power absorption for the wind, as this is a property mostly exclusive to BESS. The response is different between strategies one and two as strategy one shows an extremely unrealistic and dangerous response, not taking into account NMP dynamics. This would likely lead to the wind turbines becoming unstable if strategy one was implemented in a real a DVPP.

4.12 Hydro response

Hydro response is almost identical between the two strategies, due to its self governing nature and large power contribution. It also correctly shows NMP characteristics, while also slowly increasing power over the 300 second duration. This is likely due to an incredibly small deviation from the nominal frequency, and as hydro handles the slow return to nominal, it slowly increases its power until this deviation is resolved.

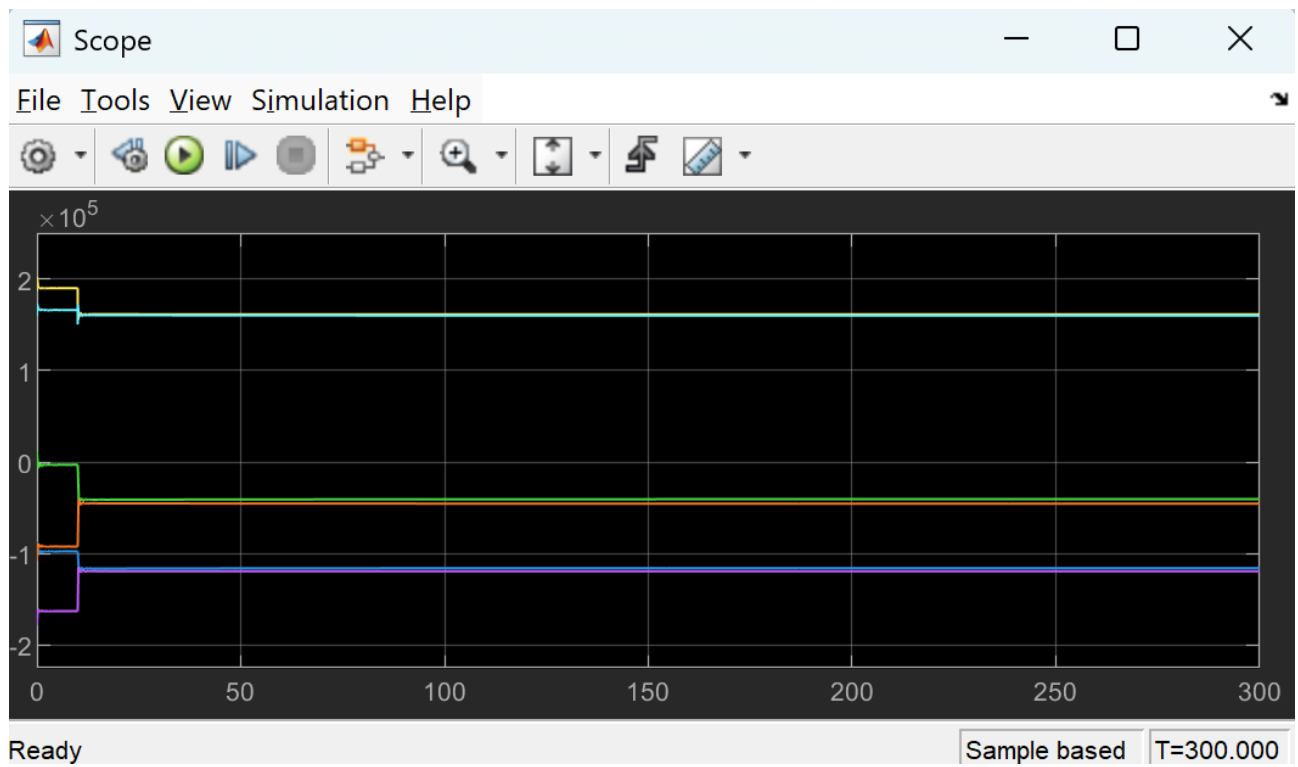


Figure 15: Voltage Strategy 1

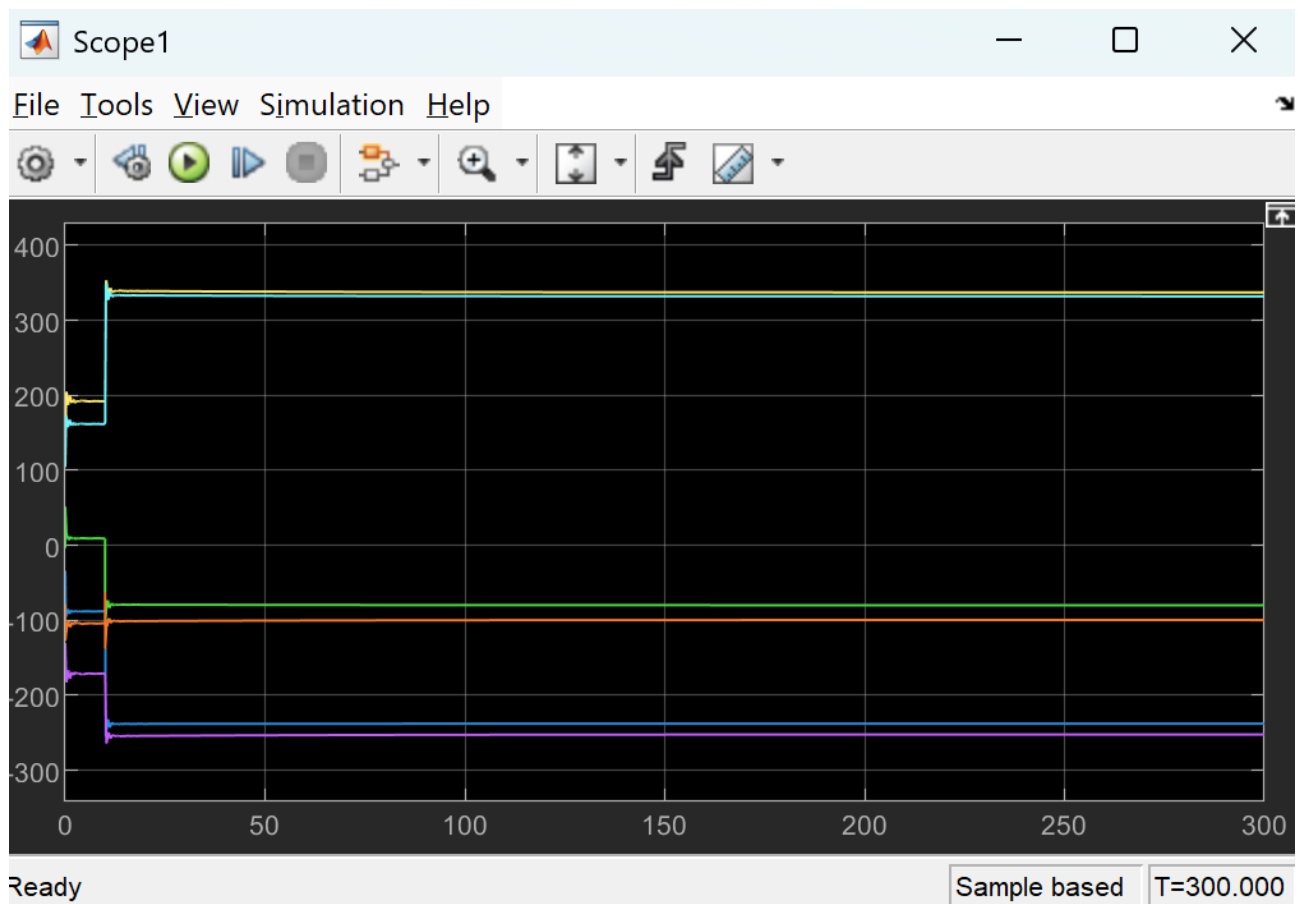


Figure 16: Current Strategy 1

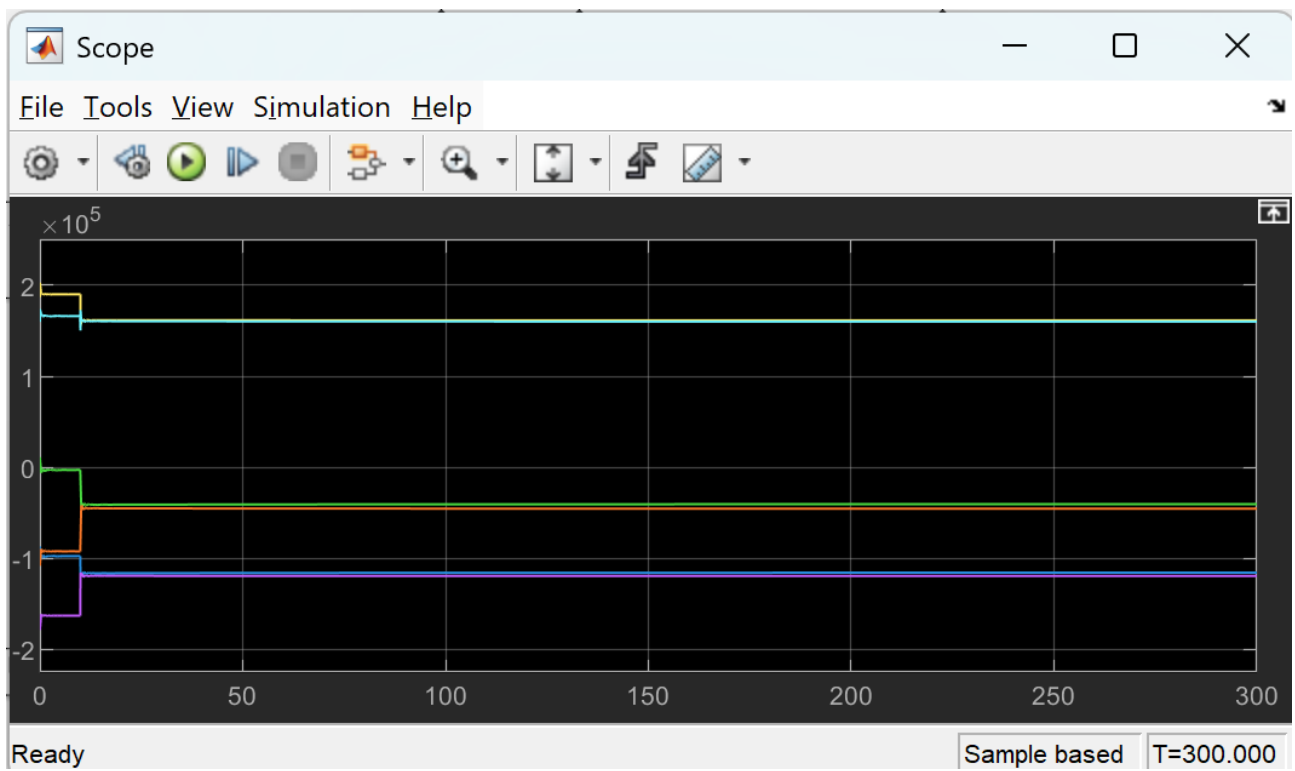


Figure 17: Voltage Strategy 2

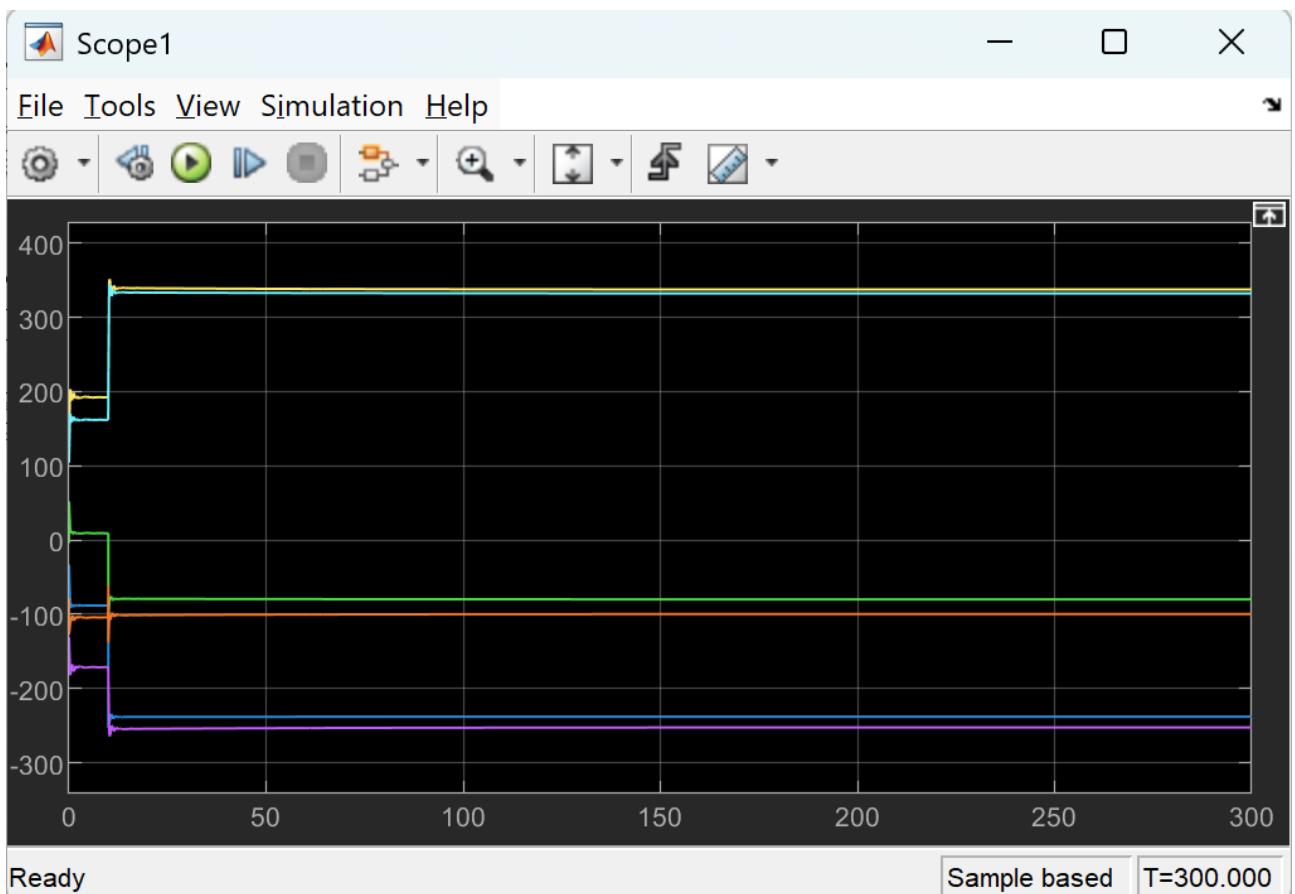


Figure 18: Current Strategy 2

5 Model

6 Discussion

7 Bibliography

References

- [1] Climate Action Tracker, *CAT net zero target evaluations*, Last updated 27 October 2025. Accessed: May 2026, Oct. 2025. [Online]. Available: <https://climateactiontracker.org/global/cat-net-zero-target-evaluations/>.
- [2] R. Tonkoski, L. A. C. Lopes, and T. H. M. El-Fouly, “Coordinated active power curtailment of grid connected PV inverters for overvoltage prevention,” *IEEE Transactions on Sustainable Energy*, vol. 2, no. 2, pp. 139–147, Apr. 2011. DOI: [10.1109/TSTE.2010.2098483](https://doi.org/10.1109/TSTE.2010.2098483).
- [3] F. Milano, F. Dörfler, G. Hug, D. J. Hill, and G. Verbič, “Foundations and challenges of low-inertia systems (invited paper),” in *2018 Power Systems Computation Conference (PSCC)*, Dublin, Ireland, Jun. 2018, pp. 1–25. DOI: [10.23919/PSCC.2018.8450880](https://doi.org/10.23919/PSCC.2018.8450880).
- [4] D. E. Ochoa, F. Galarza-Jimenez, F. Wilches-Bernal, D. A. Schoenwald, and J. I. Poveda, “Control systems for low-inertia power grids: A survey on virtual power plants,” *IEEE Access*, vol. 11, pp. 20 560–20 581, 2023. DOI: [10.1109/ACCESS.2023.3249151](https://doi.org/10.1109/ACCESS.2023.3249151).
- [5] D. Pudjianto, C. Ramsay, and G. Strbac, “Virtual power plant and system integration of distributed energy resources,” *IET Renewable Power Generation*, vol. 1, no. 1, pp. 10–16, 2007. DOI: [10.1049/iet-rpg:20060023](https://doi.org/10.1049/iet-rpg:20060023).
- [6] E. Dall’Anese, S. S. Guggilam, A. Simonetto, Y. C. Chen, and S. V. Dhople, “Optimal regulation of virtual power plants,” *IEEE Transactions on Power Systems*, vol. 33, no. 2, pp. 1868–1881, Mar. 2018. DOI: [10.1109/TPWRS.2017.2741920](https://doi.org/10.1109/TPWRS.2017.2741920).
- [7] W. Zhong, J. Chen, M. Liu, M. A. A. Murad, and F. Milano, “Coordinated control of virtual power plants to improve power system short-term dynamics,” *Energies*, vol. 14, no. 4, p. 1182, 2021. DOI: [10.3390/en14041182](https://doi.org/10.3390/en14041182).
- [8] K. Dielmann and A. van der Velden, “Virtual power plants (VPP) – A new perspective for energy generation?” In *Proceedings of the 9th International Scientific and Practical Conference of Students, Post-graduates Modern Techniques and Technologies (MTT 2003)*, Tomsk, Russia, 2003, pp. 18–20. DOI: [10.1109/SPCMTT.2003.1438108](https://doi.org/10.1109/SPCMTT.2003.1438108).
- [9] Scopus, *Search results analysis: “virtual power plant”*, Scopus database search on TITLE-ABS-KEY("virtual power plant"). Accessed: 1st of May 2025, 2024. [Online]. Available: <https://www.scopus.com/term/analyzer.uri?s=TITLE-ABS-KEY%28%22virtual+power+plant%22%29>.
- [10] Scopus, *Search results analysis: “dynamic virtual power plant”*, Scopus database search on TITLE-ABS-KEY("dynamic virtual power plant"). Accessed: 1st of May 2025, 2024. [Online]. Available: <https://www.scopus.com/term/analyzer.uri?s=TITLE-ABS-KEY%28%22dynamic+virtual+power+plant%22%29>.
- [11] A. Nieße, M. Tröschel, and H.-J. Appelrath, “Market-based self-organized provision of active power and ancillary services: An agent-based approach for smart distribution grids,” in *2012 Complexity in Engineering (COMPENG) Proceedings*, Aachen, Germany, 2012, pp. 1–5. DOI: [10.1109/CompEng.2012.6242953](https://doi.org/10.1109/CompEng.2012.6242953).
- [12] B. Marinescu, O. Gomis-Bellmunt, F. Dörfler, H. Schulte, and L. Sigrist, “Dynamic virtual power plant: A new concept for grid integration of renewable energy sources,” *IEEE Access*, vol. 10, pp. 104 980–104 995, 2022. DOI: [10.1109/ACCESS.2022.3205731](https://doi.org/10.1109/ACCESS.2022.3205731).
- [13] V. Häberle, M. W. Fisher, E. Prieto-Araujo, and F. Dörfler, “Control design of dynamic virtual power plants: An adaptive divide-and-conquer approach,” *IEEE Transactions on Power Systems*, vol. 37, no. 5, pp. 4040–4053, Sep. 2022. DOI: [10.1109/TPWRS.2021.3139775](https://doi.org/10.1109/TPWRS.2021.3139775).
- [14] J. Björk, K. H. Johansson, and F. Dörfler, “Dynamic virtual power plant design for fast frequency reserves: Coordinating hydro and wind,” *IEEE Transactions on Control of Network Systems*, vol. 10, no. 3, pp. 1266–1278, Sep. 2023. DOI: [10.1109/TCNS.2022.3181553](https://doi.org/10.1109/TCNS.2022.3181553).
- [15] J. Björk, D. V. Pombo, and K. H. Johansson, “Variable-speed wind turbine control designed for coordinated fast frequency reserves,” *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1471–1481, Mar. 2022. DOI: [10.1109/TPWRS.2021.3104905](https://doi.org/10.1109/TPWRS.2021.3104905).

- [16] M. E. Adabi and B. Marinescu, *Direct participation of dynamic virtual power plants in secondary frequency control*, 2022. arXiv: [2203.10406](https://arxiv.org/abs/2203.10406) [eess.SY]. [Online]. Available: <https://arxiv.org/abs/2203.10406>.
- [17] H. Golpîra and B. Marinescu, “Enhanced frequency regulation scheme: An online paradigm for dynamic virtual power plant integration,” *IEEE Transactions on Power Systems*, vol. 39, no. 6, pp. 7227–7239, Nov. 2024. DOI: [10.1109/TPWRS.2024.3368796](https://doi.org/10.1109/TPWRS.2024.3368796).
- [18] Y. Wang, X. Yu, and M. Jalili, “GeoDistNet: An open-source tool for synthetic distribution network generation,” *arXiv preprint arXiv:2603.29523*, 2026. DOI: [10.48550/arXiv.2603.29523](https://doi.org/10.48550/arXiv.2603.29523).
- [19] V. Häberle, A. Tayyebi, X. He, E. Prieto-Araujo, and F. Dörfler, “Grid-forming and spatially distributed control design of dynamic virtual power plants,” *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1761–1777, Mar. 2024. DOI: [10.1109/TSG.2023.3311481](https://doi.org/10.1109/TSG.2023.3311481).
- [20] Australian Energy Market Operator (AEMO), “Grid-forming BESS connections,” Australian Energy Market Operator, Fact Sheet, Dec. 2022. [Online]. Available: <https://aemo.com.au/-/media/files/initiatives/engineeringframework/2021/application-of-advanced-grid-scale-inverters-in-the-nem.pdf>.
- [21] A. Tayyebi, D. Groß, A. Anta, F. Kupzog, and F. Dörfler, “Frequency stability of synchronous machines and grid-forming power converters,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 8, no. 2, pp. 1004–1018, Jun. 2020. DOI: [10.1109/JESTPE.2020.2966524](https://doi.org/10.1109/JESTPE.2020.2966524).
- [22] D.-I. Stroe, V. Knap, M. Swierczynski, A.-I. Stroe, and R. Teodorescu, “Operation of a grid-connected lithium-ion battery energy storage system for primary frequency regulation: A battery lifetime perspective,” *IEEE Transactions on Industry Applications*, vol. 53, no. 1, pp. 430–438, 2017. DOI: [10.1109/TIA.2016.2616319](https://doi.org/10.1109/TIA.2016.2616319).
- [23] N. A. Arif, S. Mekhilef, M. Seyedmahmoudian, and A. Stojcevski, “Degradation in li-ion batteries and capacity decline under different cycling conditions: A comprehensive review,” *Journal of Energy Storage*, vol. 154, p. 121 270, 2026, ISSN: 2352-152X. DOI: [10.1016/j.est.2026.121270](https://doi.org/10.1016/j.est.2026.121270). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352152X26009345>.